



Report 4

Network Design Using Core, Distribution, and Access Layers

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**NETWORK SECURITY –
BATCH 2**



Introduction:

The objective of this task was to design and implement a hierarchical enterprise network using the Core, Distribution, and Access Layer architecture in Cisco Packet Tracer. Three routers were configured as the Core Layer and connected using OSPF to provide dynamic routing. Switches were used to connect end devices and servers, creating a scalable and reliable enterprise network. The task also focused on understanding the purpose of each network layer and verifying end-to-end connectivity.

Objectives:

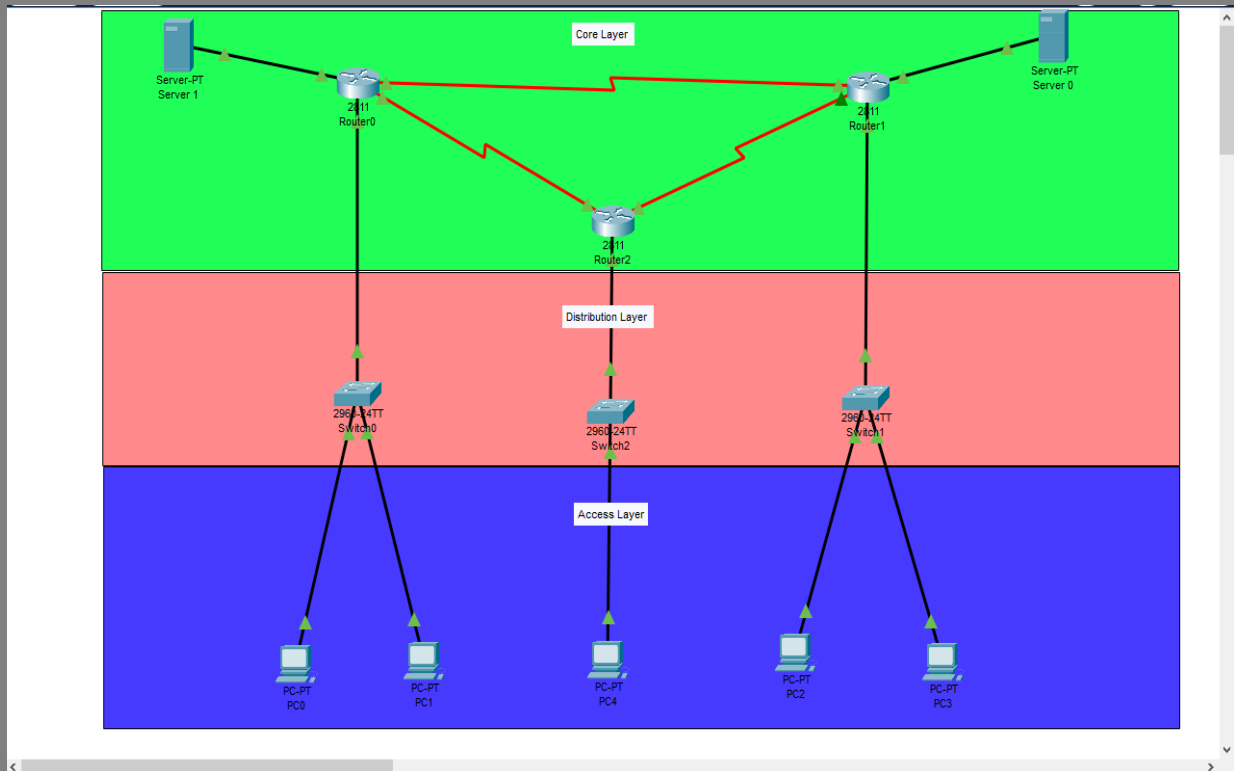
- Design a hierarchical enterprise network.
- Configure routers using OSPF dynamic routing.
- Connect switches and end devices.
- Add servers to provide enterprise network services.
- Verify routing and network connectivity.
- Understand the purpose of Core, Distribution, and Access layers.

Software and Tools Used:

- Cisco Packet Tracer
- Cisco 2811 Routers
- Cisco 2960 Switches
- PCs
- Server-PT
- OSPF Routing Protocol

Network Topology:

The network consists of three routers connected in a triangular topology using OSPF. Each router connects to a switch, and each switch connects to client PCs. Two servers were added to provide centralized network services.



Core Layer Configuration:

Three Cisco 2811 routers were configured as the Core Layer. OSPF Area 0 was implemented to exchange routing information dynamically between the routers.

Show ip int brief Router 0:

```
Router#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.10.1    YES NVRAM  up          up
FastEthernet0/1 192.168.14.1    YES manual up          up
Serial0/0/0     11.0.0.1        YES manual up          up
Serial0/0/1     unassigned      YES unset  down        down
Serial0/3/0     10.0.0.1        YES NVRAM  up          up
Vlan1          unassigned      YES NVRAM  administratively down down
Router#
```

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OSPF Neighbor Verification Router 0:

```
vlan1 unassigned 125 MVRAM administratively down down
Router#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address      Interface
2.2.2.2          0    FULL/ -         00:00:34    10.0.0.2     Serial0/3/0
Router#
```

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The output confirms that all routers successfully formed OSPF neighbor relationships, allowing routing information to be exchanged across the network.

Routing Table Router 0:

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
 C    10.0.0.0/8 is directly connected, Serial0/3/0
 L    10.0.0.1/32 is directly connected, Serial0/3/0
 C    10.0.0.2/32 is directly connected, Serial0/3/0
 11.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
 C    11.0.0.0/8 is directly connected, Serial0/0/0
 L    11.0.0.1/32 is directly connected, Serial0/0/0
 C    11.0.0.2/32 is directly connected, Serial0/0/0
 O    12.0.0.0/8 [110/200] via 10.0.0.2, 00:22:46, Serial0/3/0
 C    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
 C    192.168.10.0/24 is directly connected, FastEthernet0/0
 L    192.168.10.1/32 is directly connected, FastEthernet0/0
--More--
```

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The routing table displays directly connected and OSPF-learned routes. This verifies that the routers have successfully learned remote networks

Show ip Protocols Router 0:

```
Router#show ip protocols

Routing Protocol is "ospf 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.3 area 0
    192.168.10.0 0.0.0.255 area 0
    11.0.0.0 0.0.0.3 area 0
    192.168.14.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:08:49
    2.2.2.2          110          00:08:14
    3.3.3.3          110          00:21:39
  Distance: (default is 110)

Router#
```

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Show ip int brief Router 1:

```
Router#
Router#show ip int brief

Interface                IP-Address      OK? Method Status        Protocol
FastEthernet0/0          192.168.11.1    YES NVRAM  up            up
FastEthernet0/1          192.168.15.1    YES manual  up            up
Serial0/0/0               12.0.0.2        YES manual  up            up
Serial0/0/1               unassigned      YES unset   down          down
Serial0/3/0               10.0.0.2        YES NVRAM  up            up
Vlan1                    unassigned      YES NVRAM  administratively down down
Router#
```

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OSPF Neighbor Verification Router 1:

```
Vlan1                    unassigned      YES NVRAM  administratively down down
Router#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
1.1.1.1        0     FULL/ -         00:00:36   10.0.0.1       Serial0/3/0
3.3.3.3        0     FULL/ -         00:00:36   12.0.0.1       Serial0/0/0
Router#
```

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Routing Table Router 1:

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

 10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    10.0.0.0/8 is directly connected, Serial0/3/0
C    10.0.0.1/32 is directly connected, Serial0/3/0
L    10.0.0.2/32 is directly connected, Serial0/3/0
O    11.0.0.0/8 [110/200] via 10.0.0.1, 00:31:52, Serial0/3/0
 12.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Serial0/0/0
C    12.0.0.1/32 is directly connected, Serial0/0/0
L    12.0.0.2/32 is directly connected, Serial0/0/0
O    192.168.10.0/24 [110/101] via 10.0.0.1, 00:42:59, Serial0/3/0
 192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.11.0/24 is directly connected, FastEthernet0/0

Router#
Router#
```

The routing table displays directly connected and OSPF-learned routes. This verifies that the routers have successfully learned remote networks.

Show ip Protocols Router 1:

```
3.3.3.3 0 FULL/ 00.00.36 12.0.0.1 Serial0/0/0
Router#show ip protocols

Routing Protocol is "ospf 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 2.2.2.2
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    10.0.0.0 0.0.0.3 area 0
    192.168.11.0 0.0.0.255 area 0
    11.0.0.0 0.0.0.3 area 0
    12.0.0.0 0.0.0.3 area 0
    192.168.15.0 0.0.0.255 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110          00:15:12
    2.2.2.2          110          00:14:37
    3.3.3.3          110          00:28:02
  Distance: (default is 110)

Router#
```

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The output displays the configured OSPF process, router ID, and advertised networks.

Show Ip int brief Router 2:

```
Router#show ip int brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.12.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/3/0	11.0.0.2	YES	manual	up	up
Serial0/3/1	12.0.0.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

```
Router#
```

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The output confirms that all router interfaces are configured correctly and are in the up/up state.

OSPF Neighbor Verification Router 2:

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
2.2.2.2	0	FULL/ -	00:00:36	12.0.0.2	Serial0/3/1

```
Router#
```

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The output confirms that all routers successfully formed OSPF neighbor relationships, allowing routing information to be exchanged across the network.

Routing Table Router 2:

```
Router#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
O    10.0.0.0/8 [110/200] via 12.0.0.2, 00:25:39, Serial0/3/1
O    11.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    11.0.0.0/8 is directly connected, Serial0/3/0
C    11.0.0.1/32 is directly connected, Serial0/3/0
L    11.0.0.2/32 is directly connected, Serial0/3/0
O    12.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    12.0.0.0/8 is directly connected, Serial0/3/1
L    12.0.0.1/32 is directly connected, Serial0/3/1
C    12.0.0.2/32 is directly connected, Serial0/3/1
O    192.168.10.0/24 [110/201] via 12.0.0.2, 00:25:39, Serial0/3/1
O    192.168.11.0/24 [110/101] via 12.0.0.2, 00:25:39, Serial0/3/1
O    192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.12.0/24 is directly connected, FastEthernet0/0
L    192.168.12.1/32 is directly connected, FastEthernet0/0
O    192.168.14.0/24 [110/201] via 12.0.0.2, 00:12:46, Serial0/3/1
O    192.168.15.0/24 [110/101] via 12.0.0.2, 00:12:11, Serial0/3/1
```

```
Router#
```

The routing table displays directly connected and OSPF-learned routes. This verifies that the routers have successfully learned remote networks.

Show IP Protocols Router 2:

```
Router#show ip protocols

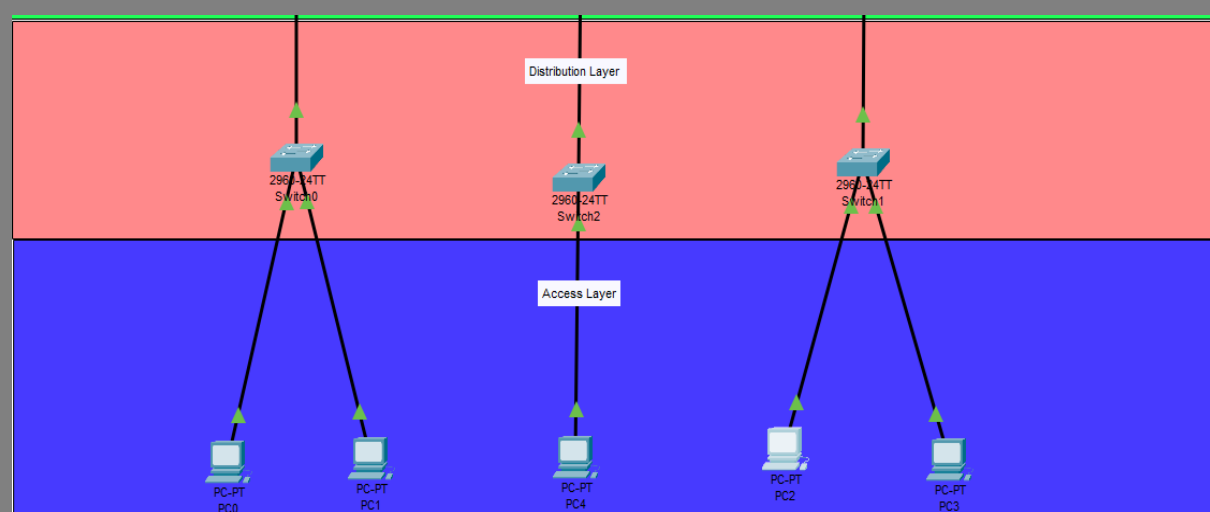
Routing Protocol is "ospf 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.3
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.12.0 0.0.0.255 area 0
    10.0.0.0 0.0.0.3 area 0
    12.0.0.0 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1          110           00:12:41
    2.2.2.2          110           00:12:06
    3.3.3.3          110           00:25:30
  Distance: (default is 110)

Router#
```

The output displays the configured OSPF process, router ID, and advertised networks.

Distribution Layer Configuration:

Each router is connected to a Cisco 2960 switch. The switches aggregate traffic from end devices and provide connectivity between the Core Layer and user devices.

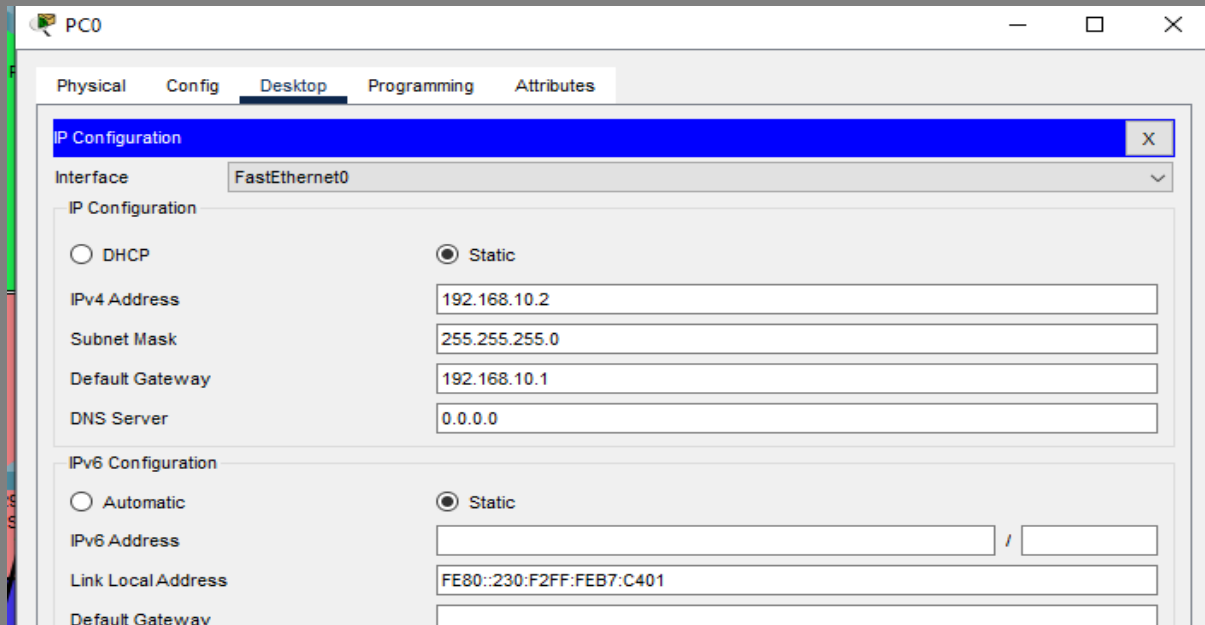


The switches act as the Distribution Layer by connecting routers to end devices and forwarding traffic within the local network.

End Devices and Servers:

Client PCs and servers were configured with static IP addresses and default gateways. Servers provide centralized network services that can be accessed by clients across different networks.

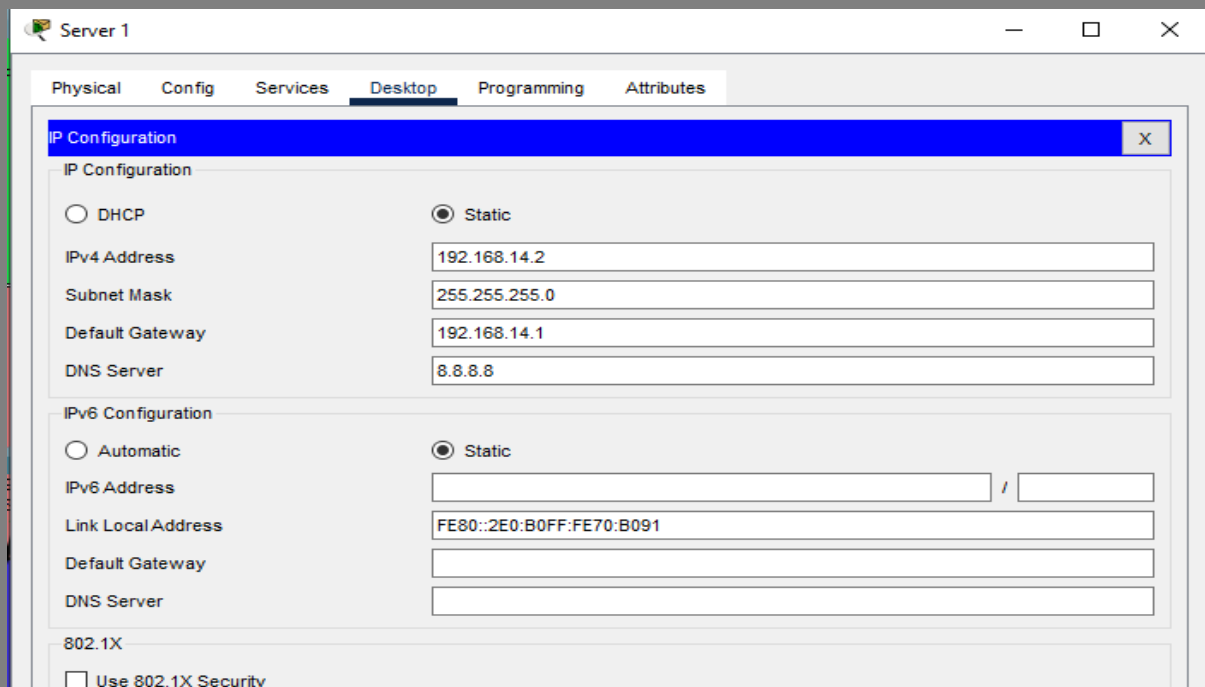
PC IP Configuration



The screenshot shows the 'PC0' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing settings for the 'FastEthernet0' interface. The 'Static' radio button is selected for both IPv4 and IPv6 configurations.

Field	Value
Interface	FastEthernet0
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	0.0.0.0
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::230:F2FF:FE77:C401
Default Gateway	

Server IP Configuration:



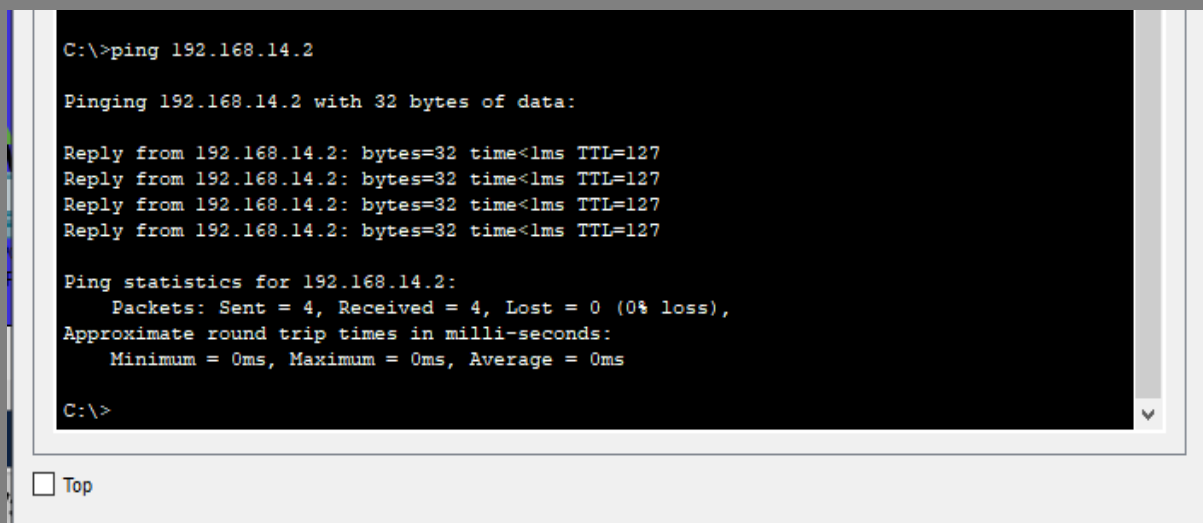
The screenshot shows the 'Server 1' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing settings for the interface. The 'Static' radio button is selected for both IPv4 and IPv6 configurations.

Field	Value
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	192.168.14.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.14.1
DNS Server	8.8.8.8
IPv6 Configuration	
☐ Automatic	☒ Static
IPv6 Address	
Link Local Address	FE80::2E0:B0FF:FE70:B091
Default Gateway	
DNS Server	
802.1X	
☐ Use 802.1X Security	

The screenshot shows the IP configuration of the client PCs and servers, including IP address, subnet mask, and default gateway.

Ping Tests:

PC 0 To Server Ping:



```
C:\>ping 192.168.14.2

Pinging 192.168.14.2 with 32 bytes of data:

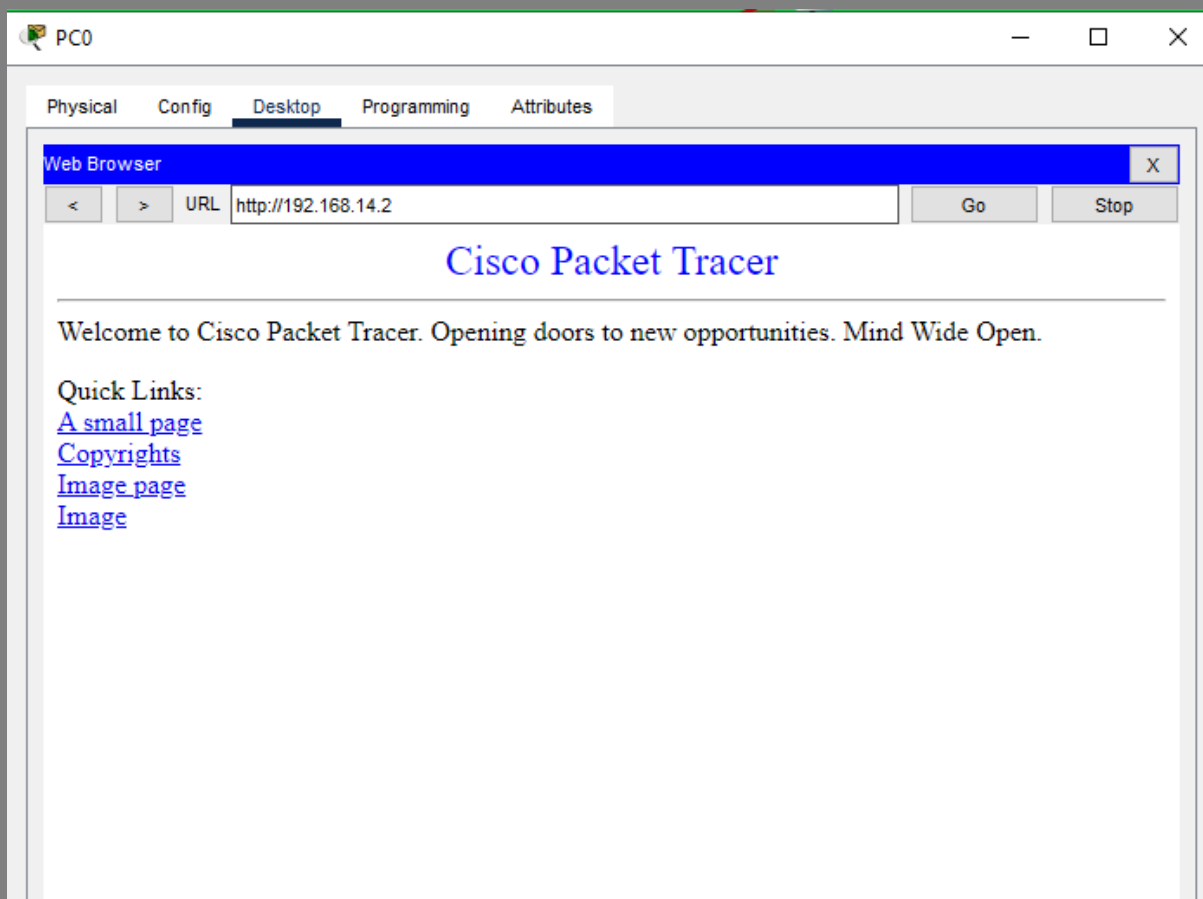
Reply from 192.168.14.2: bytes=32 time<1ms TTL=127
Reply from 192.168.14.2: bytes=32 time<1ms TTL=127
Reply from 192.168.14.2: bytes=32 time<1ms TTL=127
Reply from 192.168.14.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.14.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

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PC 0 Accessing Web Services from Server:



The screenshot confirms that client devices can successfully access the servers through the routed network.

PC 2 To Server 1 & 2:

```
C:\>ping 192.168.14.2

Pinging 192.168.14.2 with 32 bytes of data:

Reply from 192.168.14.2: bytes=32 time=30ms TTL=126
Reply from 192.168.14.2: bytes=32 time=24ms TTL=126
Reply from 192.168.14.2: bytes=32 time=19ms TTL=126
Reply from 192.168.14.2: bytes=32 time=29ms TTL=126

Ping statistics for 192.168.14.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 19ms, Maximum = 30ms, Average = 25ms

C:\>ping 192.168.15.2

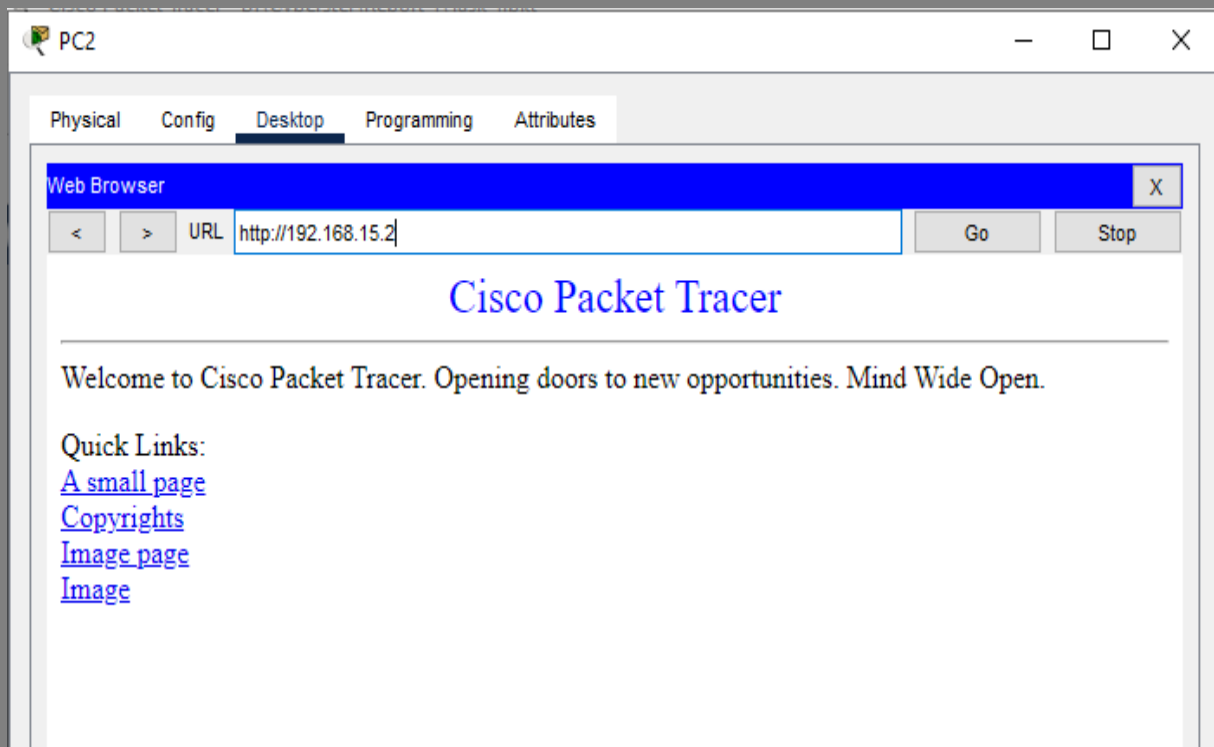
Pinging 192.168.15.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.15.2: bytes=32 time<1ms TTL=127
Reply from 192.168.15.2: bytes=32 time<1ms TTL=127
Reply from 192.168.15.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.15.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

PC 2 Accessing Web Services From Server 2:



Layer Explanation:

Core Layer:

The Core Layer consists of the three routers. It provides the backbone of the network by forwarding traffic between different networks using OSPF. It ensures high-speed, reliable communication across the enterprise.

Distribution Layer:

The Distribution Layer consists of the Cisco 2960 switches. These switches connect the Core Layer to end devices, aggregate network traffic, and provide connectivity within each local network.

Access Layer:

The Access Layer includes the end devices such as PCs and servers. These devices access network resources through the switches and communicate with other networks via the routers.

Difference Between Access Layer and Distribution Layer:

Access Layer	Distribution Layer
Connects end devices such as PCs and servers.	Connects the Access Layer to the Core Layer.
Provides user access to the network.	Aggregates traffic from multiple access networks.
Used by end users.	Used for traffic forwarding and policy implementation
Located at the edge of the network	Located between the Core and Access Layers.

Learning Outcomes:

Through this task, I learned how to design a hierarchical enterprise network using Core, Distribution, and Access layers. I gained practical experience in configuring OSPF, assigning IP addresses, verifying routing information, and testing end-to-end connectivity. I also understood the role of each network layer in building scalable and reliable enterprise networks.

Conclusion:

The enterprise network was successfully designed and implemented using Cisco Packet Tracer. OSPF dynamic routing enabled communication between all networks, and switches provided connectivity for end devices and servers. The task improved my understanding of hierarchical network architecture and enterprise networking concepts.